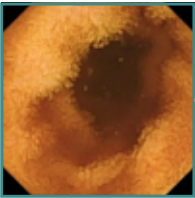


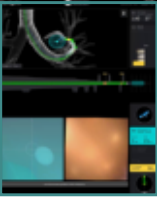
Lane A: passive-video forecasting
evaluated across three domains



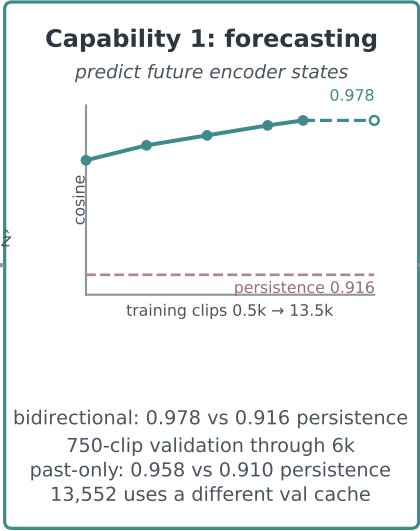
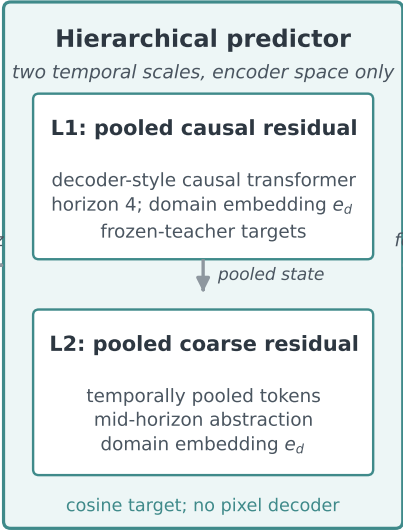
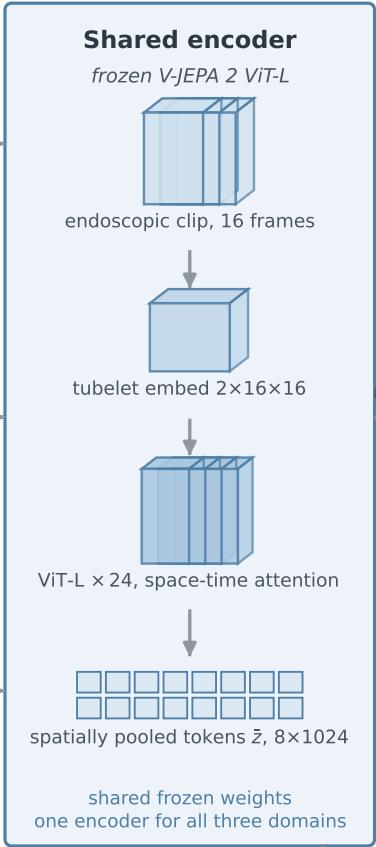
CholecT50 (laparoscopy)
last observed frame



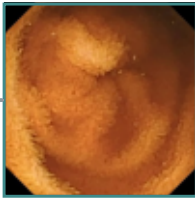
Kvasir-Capsule (GI)
last observed frame



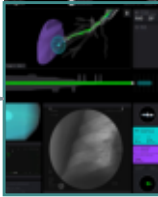
ION (bronchoscopy)
navigation-console capture



CholecT50 (laparoscopy)
predicted-latent retrieval



Kvasir-Capsule (GI)
predicted-latent retrieval



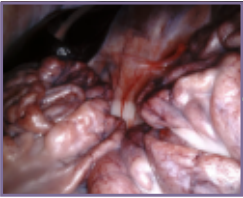
ION (bronchoscopy)
predicted-latent retrieval

loss-of-view warning

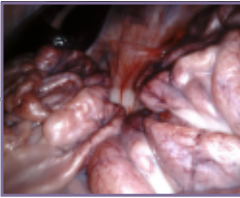
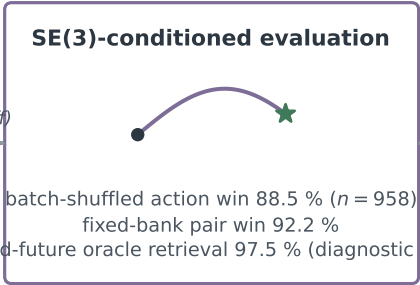
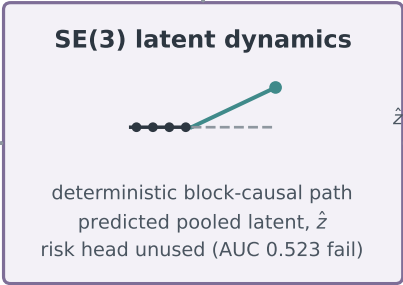
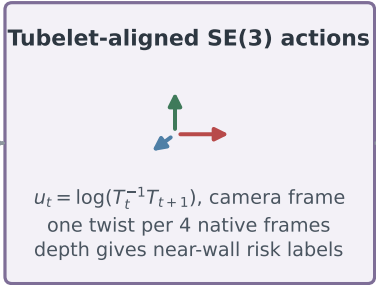
camera-handling training

cross-domain representation

Lane B: physical grounding
audited on pose-gated data only



SCARED (laparoscopy)
frame + calibrated pose



SCARED (laparoscopy)
SE(3)-conditioned retrieval

same encoder weights
protocol-specific tokens

twist u

$\hat{z}$ (risk off)

retrieval proxy

Correctness audits across the study

Input-sensitivity tests
history/action; domain where present

C3VD depth-warp diagnostic
convention not validated

Action negatives
batch shuffle / fixed bank

Sequence-grouped splits
STIR patient grouping; logged audit contacts